

Localization of D-instability in star interconnections and a sharp bound on synchronized groups

September 2026

Abstract

A real square matrix A is D-stable if AD is Hurwitz for every positive diagonal matrix D , that is, if the linear system it describes is stable whatever the positive time scales of its coordinates. We study star interconnections: a core matrix $B \in \mathbb{R}^{n \times n}$ together with first-order attachments that read the core through rank-one interfaces $u_c v_c^\top$, grouped into k channels, with nonzero feedback loads of either sign. Assigning each attachment to be fast (quasi-steady), slow (frozen), or a member of a single synchronized group of its channel defines a *boundary system*: a star of dimension $n + k$ with at most one dynamic lump per channel. We prove that every eigenvalue in the closed right half-plane of every positive scaling of the star is also an eigenvalue of some positive scaling of one of its boundary systems, in general with different time constants of the core. No hypothesis on the core is needed. Conversely, strict instability of a boundary system lifts to the star. Hence the star is D-semistable exactly when all its boundary systems are, and it is D-stable if all its boundary systems are; an explicit example shows that the last implication cannot be reversed. Neither feature of the boundary systems can be dropped: for every $k \geq 2$ a cyclic star built on the secant condition is not D-stable, while every boundary system with fewer than k synchronized groups is; and already for one channel, any number of pieces may have to be lumped into one synchronized group. The proof combines a compact parametrization of each attachment's frequency response at a fixed eigenvalue, an exact identity showing that two attachments of one channel have independent tangents unless their time constants agree, and an exit argument phrased as a maximum over a compact cube. The theorems, propositions and lemmas are formally verified in Lean 4 with Mathlib.

Keywords: D-stability; diagonal scaling; time scales; quasi-steady-state approximation; lumping; model reduction; secant condition; formal verification.

MSC 2020: 15A18, 93D09 (primary); 34D20, 93C15, 93C70 (secondary).

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1 Introduction

A real $N \times N$ matrix A is *D-stable* if AD has all its eigenvalues in the open left half-plane for every positive diagonal matrix D [15, 5]. Writing $T = D^{-1}$, the eigenvalues of AD are the roots of $\det(zT - A)$, the characteristic values of the system $T\dot{x} = Ax$ in which coordinate i has time constant T_{ii} . D-stability is therefore stability for all positive time scales, a natural requirement whenever capacities, volumes or turnover rates are uncertain; see the surveys [18, 23].

In applications such systems are rarely analysed at full size. One eliminates fast species by a quasi-steady-state approximation, freezes slow ones [25, 22], and lumps species with a common time constant into one pool [33]. The first two operations are limits in the space of time scales; the third restricts to a diagonal of it. Stability of a reduced model says nothing, in general, about the full family: a quasi-steady reduction can lose an oscillation that the full model has [16, 21]. This paper asks which reduced models decide D-stability, and how many time scales a reduced model must retain, for a class of interconnections in which the question has a sharp answer.

Stars. A *star*¹ consists of a core $B \in \mathbb{R}^{n \times n}$ and m first-order attachments (*pieces*). Piece p belongs to a channel $\text{ch}(p)$ among k channels. Channel c reads the core through $v_c^\top x$ and feeds back along u_c , with $u_c, v_c \in \mathbb{R}^n$. Piece p has a nonzero load r_p of either sign:

$$\dot{x} = Bx + \sum_p r_p u_{\text{ch}(p)} w_p, \quad \dot{w}_p = v_{\text{ch}(p)}^\top x - w_p,$$

with all coordinates then given arbitrary positive time constants. Binding sites, decoys and buffers acting on one species are coordinate channels, $u_c = v_c = \mathbf{e}_i$; a complex formed from several species, or a sensor reading a weighted sum, gives a general rank-one channel.

Boundary systems. A *boundary choice* sends every piece to F (fast, quasi-steady: it becomes the static load $r_p u_{\text{ch}(p)} v_{\text{ch}(p)}^\top$ of the core), S (slow, frozen: it disappears), or D (it joins the synchronized group of its channel, whose pieces share one time constant and act as a single lumped pool). The resulting *boundary system* is again a star, with at most one dynamic lump per channel (Definition 2.2). It is the reduced model obtained by quasi-steady elimination, freezing, and lumping within a channel.

Results. We prove the following, for arbitrary cores, arbitrary rank-one interfaces and nonzero loads of either sign.

- (A) *Spectral localization* (Theorem 2.3). If some positive scaling of the star has an eigenvalue z with $\text{Re } z \geq 0$, then some positive scaling of some boundary system has the same eigenvalue z , although in general not with the same core time constants. No stability hypothesis on the core is used.

¹The name refers to the interconnection pattern of a core with attachments; it is unrelated to star graphs and star sign patterns in combinatorial matrix theory.

- (B) *Lifting* (Theorem 2.4). If some scaling of a boundary system has an eigenvalue with positive real part, so does some scaling of the star (in general at a different eigenvalue).
- (C) *Criterion* (Corollary 2.5). The star is D-semistable if and only if all its boundary systems are. It is D-stable if all its boundary systems are, and Proposition 2.6 shows that this implication is not an equivalence.
- (D) *Sharpness* (Theorem 2.7 and Proposition 2.8). For every $k \geq 2$ there is a k -channel star that is not D-stable although every boundary system with fewer than k synchronized groups is D-stable. For $k = 1$ and every $m \geq 2$ there is a star with m pieces that is not D-stable although every boundary system in which fewer than m pieces are synchronized is D-stable. So neither the number of groups nor their size can be bounded in general.

In modelling terms: for star interconnections, screening stability under all time scales by reduced models is exact for D-semistability, provided one checks every combination of quasi-steady, frozen and lumped pieces with one lumped pool per interface; and one can do neither with fewer lumped pools nor with unlumped pieces. The number of boundary choices is exponential in the number of pieces. Some such growth is to be expected: for binary-encoded loads, deciding D-stability is coNP-hard already for stars with one coordinate channel and a 4×4 core whose entries depend on the instance [14].

Ideas of the proof. Eliminating the pieces at an eigenvalue z with $\operatorname{Re} z \geq 0$ leaves an $n \times n$ pencil whose determinant is affine in the value ℓ_c of each channel (Lemmas 3.1 and 3.2). For non-real z , the value $r_p/(1 + z t_p)$ of one piece runs over an arc as its time constant t_p runs over $(0, \infty)$. We parametrize this arc by $s \in [0, 1]$, with $s = 0$ the quasi-steady end and $s = 1$ the frozen end, so that the configuration space becomes the compact cube $[0, 1]^P$. An exact identity (Lemma 4.2) shows that two pieces of one channel move the channel value in independent directions unless their interior parameters agree. We then dilate the core time constants, which pushes the contact out to a maximal dilation over the compact set of contacts; at the maximum, openness forbids two unequal interior parameters in the channel, so the channel is of boundary type (Lemma 5.1). Treating the channels one by one gives Theorem 2.3 for non-real z ; real eigenvalues are handled by an explicit sign-class construction. Lifting is Hurwitz's theorem on zeros of uniform limits. Sharpness uses the secant condition [31, 30, 4]: a fast channel removes one stage from a negative feedback cycle, a slow channel cuts it, and the thresholds $\sec^N(\pi/N)$ decrease strictly in N .

Related work. Johnson [19] characterized D-stability of a stable matrix by the nonsingularity of $A \pm iD$, and Kushel [24] expands $\det(A + iD)$ recursively, eliminating one scaling variable at a time; this is exact at the first step and a hierarchy of sufficient tests beyond. Our elimination removes whole attachment modules instead, and the reduced dimension $n + k$ does not grow with the number of pieces. Hartfiel [17] showed that the set of D-stable matrices is not open and gave a necessary condition for a matrix to lie in its interior (robust D-stability), in terms of principal submatrices of A and their inverses; Cain [10] showed that, for a D-stable A , conditions of this kind characterize robust D-stability; see the survey [23]. In time-scale language, principal submatrices and their inverses arise in limits in which coordinates of A become frozen or quasi-static. Boundary systems take such limits only on the attachments, and they keep dynamic lumped pools, which Theorem 2.7 and Proposition 2.8 show to be indispensable. The link between D-stability and many small time constants goes back to Khalil and Kokotović [20], Abed [1, 2], and Abed and Tits [3], who gave conditions of D-stability type on the fast subsystem (block D-stability, and strong D-stability, the robust form of D-stability) that ensure stability of multiparameter singularly perturbed systems when the parasitic time constants are small; here, in contrast, all time scales in $(0, \infty)$ are free and the

localization is exact. Berman and Hershkowitz [7] characterized acyclic D-stable matrices, and Simpson-Porco and Monshizadeh [26] gave a necessary and sufficient condition for diagonal stability of a class of matrices with a rank-one interconnection structure. For a single negative feedback cycle, the secant condition [31, 30, 27, 4] gives stability for all time constants below the threshold $\sec^N(\pi/N)$ and instability at equal time constants above it; Blanchini, Cuba Samaniego, Franco and Giordano [9] observed that equal time constants are extremal and that a sufficiently slow element stabilizes a loop. Our sharpness constructions read these facts through boundary systems. Vertex results for multiaffine families on bounded boxes [8] have a similar flavour: Theorem 2.3 shows that, for the piece time constants, it suffices to consider configurations in which the pieces of each channel take only the values 0, ∞ and one common finite value, while the core time constants remain free. Inheritance results for reaction networks [6, 32] transfer instability upward, as in Theorem 2.4; to our knowledge the downward direction, Theorem 2.3, has no counterpart there. For one coordinate channel with positive loads, a D-stable core and $n \geq 2$, a companion paper [12] gives an exact test for D-stability by the boundary systems with at most one dynamic piece, in which the purely static system enters only through the sign of its determinant. The present results hold for any core, any interfaces, any signs and any number of channels; in this generality the criterion is exact for D-semistability, while for D-stability it is only sufficient (Proposition 2.6), and single dynamic pieces no longer suffice (Proposition 2.8). To summarize the relation to the literature: the secant inequality and Hurwitz’s theorem, which we use for sharpness and lifting, are classical, and so is the observation that a quasi-steady reduction can lose an oscillation. What is new here, to our knowledge, is the downward direction (Theorem 2.3), with no hypothesis on the core, together with the proof that neither the number nor the size of the synchronized groups it needs can be reduced.

Formal verification. Every theorem, proposition, corollary and lemma of this paper is formally verified in Lean 4 [13] with Mathlib [29], as stated; Section 9 lists the declarations. The proofs below follow the formal ones. What the formalization certifies is the Lean statements; Section 9 explains how they match the definitions and statements printed here, and which parts of the text (remarks, the modelling reading, the observations in the Discussion and the complexity statements quoted from [14]) are not formalized.

Organization. Section 2 fixes notation and states the results. Section 3 eliminates the pieces. Section 4 studies one channel at a fixed eigenvalue, and Section 5 proves localization of non-real eigenvalues. Section 6 treats real eigenvalues and proves lifting, the criterion and Proposition 2.6. Section 7 proves the sharpness results. Section 8 discusses scope and open problems, and Section 9 the formalization.

2 Setting and main results

2.1 D-stability

Let $A \in \mathbb{R}^{N \times N}$. For a vector $d \in (0, \infty)^N$ write $AD = A \operatorname{diag}(d)$, a *positive scaling* of A . We say that A is

- *D-stable* if every positive scaling of A has all eigenvalues in $\{\operatorname{Re} z < 0\}$;
- *D-semistable* if no positive scaling of A has an eigenvalue in $\{\operatorname{Re} z > 0\}$;
- *D-unstable* if it is not D-semistable, and *marginal* if it is D-semistable but not D-stable, that is, some positive scaling has an eigenvalue on the imaginary axis (possibly 0) and none has one in $\{\operatorname{Re} z > 0\}$.

Table 1: Modelling reading of the objects (interpretation only).

Object	Reading
positive scaling AD	a choice of positive time constants $T = D^{-1}$ (capacities, volumes, turnover)
coordinate channel $u_c = v_c = \mathbf{e}_i$	attachments binding and releasing one species (sites, decoys, buffers)
general rank-one channel	an attachment driven by $v_c^\top x$ and acting along u_c (a complex, a sensor)
piece role F	quasi-steady-state approximation of that piece
piece role S	the piece is frozen on the time scale of interest
synchronized group D	pieces of one channel with a common time constant, lumped into one pool
boundary system Bd_σ	one reduced model obtained by these operations (possibly at other core time constants than the full model)

Since AD and DA are similar, the side on which one scales does not matter. With $T = \text{diag}(d)^{-1}$, the *time constants*, a number z is an eigenvalue of AD if and only if $\det(zT - A) = 0$.

2.2 Stars and boundary systems

Definition 2.1 (Star). Let $B \in \mathbb{R}^{n \times n}$, let C be a finite set of k *channels* with vectors $u_c, v_c \in \mathbb{R}^n$, and let P be a finite set of m *pieces* with a channel map $\text{ch}: P \rightarrow C$ and loads $r \in \mathbb{R}^P$. The *star* $A = \text{star}(B, u, v, \text{ch}, r) \in \mathbb{R}^{(n+m) \times (n+m)}$ is

$$A = \begin{pmatrix} B & U \\ V & -I_m \end{pmatrix}, \quad U\mathbf{e}_p = r_p u_{\text{ch}(p)}, \quad \mathbf{e}_p^\top V = v_{\text{ch}(p)}^\top.$$

Throughout, the time constants of a positive scaling of a star are written $T = \text{diag}(q, t)$ with $q \in (0, \infty)^n$ (core) and $t \in (0, \infty)^P$ (pieces), and $Q = \text{diag}(q)$. For $z \in \mathbb{C}$ with $1 + zt_p \neq 0$ for all p , the *channel values* and the *reduced pencil* are

$$\ell_c(z) = \sum_{\text{ch}(p)=c} \frac{r_p}{1 + zt_p}, \quad M_q(\ell) = zQ - B - \sum_{c \in C} \ell_c u_c v_c^\top. \quad (1)$$

Definition 2.2 (Boundary systems). A *boundary choice* is a map $\sigma: P \rightarrow \{\text{F}, \text{S}, \text{D}\}$. Its *boundary system* is the star with one piece per channel

$$\text{Bd}_\sigma = \text{star}(B_\sigma, u, v, \text{id}_C, R^\sigma), \quad B_\sigma = B + \sum_{\sigma(p)=\text{F}} r_p u_{\text{ch}(p)} v_{\text{ch}(p)}^\top, \quad R_c^\sigma = \sum_{\text{ch}(p)=c, \sigma(p)=\text{D}} r_p,$$

of dimension $n + k$. The *synchronized group* of channel c is $\{p : \text{ch}(p) = c, \sigma(p) = \text{D}\}$, and the number of synchronized groups of σ is the number of channels whose group is nonempty.

The scaling of a boundary system is $\text{diag}(d)$ with entries d_c on the lump coordinates, and its time constants are written (q, τ) , $\tau \in (0, \infty)^C$. The lump of a channel whose group is empty carries load 0; the matrix is then block triangular, and the lump only contributes the eigenvalue $-d_c$. A boundary system thus has at most one dynamic lump per channel. Its reduced pencil is $zQ - B_\sigma - \sum_c R_c^\sigma (1 + z\tau_c)^{-1} u_c v_c^\top$. For $z \neq 0$ with $\text{Re } z \geq 0$ this is the limit of the pencil (1) of the star as the time constants of F-pieces tend to 0 and those of S-pieces tend to ∞ , with the pieces of the group of channel c all at time constant τ_c . Lumping is exact within a channel because its pieces share u_c and v_c . Table 1 records the modelling reading; it is interpretation only and is not used in the proofs.

Pieces of load 0 do not act on the core: deleting them only removes eigenvalues $-d_p$. The assumption of nonzero loads below is therefore a normalization.

2.3 Main results

Theorem 2.3 (Spectral localization). *Let A be a star with all loads nonzero, let $z \in \mathbb{C}$ with $\operatorname{Re} z \geq 0$, and suppose that z is an eigenvalue of some positive scaling of A . Then there is a boundary choice σ such that z is an eigenvalue of some positive scaling of Bd_σ .*

Theorem 2.4 (Lifting). *Let A be a star. If some positive scaling of some boundary system Bd_σ has an eigenvalue with positive real part, then some positive scaling of A has an eigenvalue with positive real part.*

Corollary 2.5 (Criterion). *Let A be a star with all loads nonzero.*

- (a) *If every boundary system Bd_σ is D -stable, then A is D -stable.*
- (b) *A is D -semistable if and only if every boundary system Bd_σ is D -semistable. Equivalently, A is D -unstable if and only if some Bd_σ is.*

Proposition 2.6 (The criterion is not an equivalence). *Take the core B below, one channel with $u = v = \mathbf{e}_1$, and one piece of load 1, so that the star is A :*

$$B = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

The star A is D -stable, but its boundary system with the piece fast has the eigenvalue i at unit time constants.

For sharpness we use two examples. The first is cyclic. For $k \geq 2$ index the channels by $\mathbb{Z}/k$ and let $\operatorname{Cyc}_k(\rho)$, $\rho > 0$, be the star with core $-I_k$, one piece of load ρ in every channel, $v_c = \mathbf{e}_{c+1}$, $u_c = \mathbf{e}_c$ for $c \neq k-1$, and $u_{k-1} = -\mathbf{e}_{k-1}$. Its graph is a single negative feedback cycle through $2k$ first-order stages, $x_0 \leftarrow w_0 \leftarrow x_1 \leftarrow w_1 \leftarrow \cdots \leftarrow w_{k-1} \leftarrow x_0$, with loop gain $g = \rho^k$.

Theorem 2.7 (Sharpness of the number of groups). *Let $k \geq 2$ and $\rho > 0$ with*

$$\rho \cos^2\left(\frac{\pi}{2k}\right) > 1 \quad \text{and} \quad \rho^k \cos^{2k-1}\left(\frac{\pi}{2k-1}\right) < 1. \quad (2)$$

Then at unit time constants $\operatorname{Cyc}_k(\rho)$ has the eigenvalue $\sqrt{\rho}e^{i\pi/2k} - 1$, whose real part is positive; in particular $\operatorname{Cyc}_k(\rho)$ is D -unstable. Every boundary system of $\operatorname{Cyc}_k(\rho)$ with fewer than k synchronized groups is D -stable. For every $k \geq 2$ some ρ satisfies (2).

Since both conditions in (2) are monotone in ρ , the admissible ρ form an open interval. In terms of the loop gain, (2) reads $\sec^{2k}(\pi/2k) < g < \sec^{2k-1}(\pi/(2k-1))$; for $k = 2$ this is $4 < g < 8$ (Example 7.4, Figure 2). In Cyc_k every channel carries one piece, so Theorem 2.7 concerns the number of dynamic channels. The second example shows that lumping several pieces is also necessary, and that no bound on the size of a synchronized group is possible.

Proposition 2.8 (Synchronized groups of every size are needed). *Let $m \geq 2$ and $c > 2$ with $(m-1)c^3 < 8m$; for instance $c = 2 + 1/(2m)$. Take the core*

$$B = \begin{pmatrix} -1 & 0 \\ 1 & -1 \end{pmatrix},$$

one channel with $v = \mathbf{e}_2$ and $u = -\mathbf{e}_1$, and m pieces of load c^3/m each. At unit time constants the star has the eigenvalue $-1 + ce^{i\pi/3}$, whose real part $c/2 - 1$ is positive; in particular it is D -unstable. Every boundary system in which fewer than m pieces are synchronized is D -stable. For $m = 2$ one may take $c = 5/2$: two loads $125/16$ and the eigenvalue $\frac{1}{4} + \frac{5\sqrt{3}}{4}i$.

The star of Proposition 2.8 is a negative feedback loop $x_1 \rightarrow x_2 \rightarrow \{w_1, \dots, w_m\} \dashv x_1$ of total gain $c^3 > 8$. Any $m - 1$ pools together carry at most $(m - 1)c^3/m < 8$, below the three-stage threshold $\sec^3(\pi/3) = 8$, and quasi-steady pools only add static feedback; only all m pools lumped together destabilize.

Remark 2.9 (Scope of the statements). We record three features of the statements.

- (i) Theorem 2.3 holds for $z = 0$ with $\sigma \equiv \mathbf{F}$, and for real $z > 0$ with a boundary choice that has no fast pieces (Lemmas 6.1 and 6.2); in both cases the same core time constants can be kept. For non-real z the proof of Proposition 5.2 multiplies all core time constants by a common factor $\kappa \geq 1$.
- (ii) Corollary 2.5(a) asks for D-stability of every boundary system, including the purely static ones, $\text{Bd}_{\sigma \equiv \mathbf{F}}$ and $\text{Bd}_{\sigma \equiv \mathbf{S}}$ (whose core is B). Proposition 2.6 shows that a boundary system can be marginal while the star is D-stable: the limits $t_p \rightarrow 0$ and $t_p \rightarrow \infty$ are not attained.
- (iii) Bd_σ depends on σ only through the pairs $(\sum_{\text{ch}(p)=c, \sigma(p)=\mathbf{F}} r_p, R_c^\sigma)$, $c \in C$, of fast and synchronized load sums. Distinct boundary choices therefore often give the same boundary system; when all loads of a channel are equal, only the numbers of its fast and synchronized pieces matter.

3 Elimination of the pieces

Lemma 3.1 (Elimination). *Let A be a star, $q \in (0, \infty)^n$, $t \in (0, \infty)^P$, and $z \in \mathbb{C}$ with $1 + zt_p \neq 0$ for all p . Then z is an eigenvalue of $A \text{diag}(q, t)^{-1}$ if and only if the reduced pencil $M_q(\ell(z))$ of (1) is singular.*

Proof. An eigenvector of $A \text{diag}(q, t)^{-1}$ for z is $\text{diag}(q, t)(x, w)$ with $(x, w) \neq 0$ and $(zT - A)(x, w) = 0$, $T = \text{diag}(q, t)$. The row of piece p reads $zt_p w_p = v_{\text{ch}(p)}^\top x - w_p$, so $w_p = v_{\text{ch}(p)}^\top x / (1 + zt_p)$. Substituting into the core rows $zQx = Bx + \sum_p r_p u_{\text{ch}(p)} w_p$ and grouping the pieces by channel gives $M_q(\ell(z))x = 0$. If $x = 0$ then $w = 0$, so $x \neq 0$. Conversely, a nonzero kernel vector x of $M_q(\ell(z))$ and $w_p = v_{\text{ch}(p)}^\top x / (1 + zt_p)$ satisfy $(zT - A)(x, w) = 0$. $\square$

Lemma 3.2 (Rank-one updates are affine). *For $M \in \mathbb{C}^{n \times n}$, $u, v \in \mathbb{C}^n$ and $\lambda \in \mathbb{C}$,*

$$\det(M - \lambda uv^\top) = \det M + \lambda \beta(M, u, v), \quad \beta(M, u, v) = \det \begin{pmatrix} M & u \\ v^\top & 0 \end{pmatrix}.$$

Proof. By the Schur complement of the unit $(2, 2)$ entry, $\det(M - \lambda uv^\top) = \det \begin{pmatrix} M & u \\ \lambda v^\top & 1 \end{pmatrix}$. The last row is $\lambda(v^\top, 0) + (0, 1)$, and the determinant is linear in it. $\square$

In particular, for fixed z and q , $\det M_q(\ell)$ is affine in each channel value ℓ_c separately. No invertibility of $zQ - B$ is needed.

Lemma 3.3 (Diagonal dominance). *Let $q \in (0, \infty)^n$ and $K \geq 0$, and consider channel values $\ell \in \mathbb{C}^C$ with $|\ell_c| \leq K$ for all c .*

- (a) *If $z \neq 0$, there is $\kappa_0 \geq 1$ such that $M_{\kappa q}(\ell)$ is nonsingular for all $\kappa \geq \kappa_0$ and all such ℓ .*
- (b) *There is $R_0 > 0$ such that $M_q(\ell)$ is nonsingular for all z with $|z| \geq R_0$ and all such ℓ .*

Proof. Put $h_i = \sum_j (|B_{ij}| + K \sum_c |u_{c,i}| |v_{c,j}|)$ and $E = B + \sum_c \ell_c u_c v_c^\top$, so that $\sum_j |E_{ij}| \leq h_i$ and $M_q(\ell) = zQ - E$. If $|z|q_i > h_i$ for every i , the matrix $zQ - E$ is strictly diagonally dominant by rows, hence nonsingular by the Lévy–Desplanques theorem [28]. For (a) take $\kappa_0 = 1 + \sum_i h_i / (|z|q_i)$, so that $|z|\kappa q_i \geq |z|\kappa_0 q_i > h_i$ for $\kappa \geq \kappa_0$; for (b) take $R_0 = 1 + \sum_i h_i / q_i$. $\square$

4 One channel at a fixed eigenvalue

Throughout this section and the next, $z \in \mathbb{C}$ is fixed with

$$\operatorname{Re} z \geq 0 \quad \text{and} \quad \operatorname{Im} z \neq 0.$$

For $s \in \mathbb{R}$ put $\alpha(s) = 1 - s + sz$ and

$$\operatorname{val}_z(s) = \frac{1-s}{1-s+sz}.$$

The denominator never vanishes: $\operatorname{Im} \alpha(s) = s \operatorname{Im} z$ vanishes only at $s = 0$, where $\alpha = 1$.

Lemma 4.1 (Arc parametrization). *The function val_z has the following properties.*

- (a) $\operatorname{val}_z(0) = 1$ and $\operatorname{val}_z(1) = 0$.
- (b) For $t > 0$, $1/(1+zt) = \operatorname{val}_z(t/(1+t))$; for $s \in (0, 1)$, $\operatorname{val}_z(s) = 1/(1+z \frac{s}{1-s})$.
- (c) $|\operatorname{val}_z(s)| \leq 1$ for $s \in [0, 1]$.
- (d) val_z is differentiable on $\mathbb{R}$, with $\operatorname{val}'_z(s) = -z/\alpha(s)^2$.

Proof. (a) and (b) are direct. For (c), $|1-s| = 1-s \leq 1-s+s \operatorname{Re} z = \operatorname{Re} \alpha(s) \leq |\alpha(s)|$. For (d), the quotient rule gives the numerator $-\alpha(s) - (1-s)(z-1) = -z$. $\square$

Thus, as its time constant runs over $[0, \infty]$, a piece of load r contributes to its channel the points $r \operatorname{val}_z(s)$, $s \in [0, 1]$, of an arc from r (the quasi-steady end, $s = 0$) to 0 (the frozen end, $s = 1$); see Figure 1. The whole configuration space of the pieces is the compact cube $[0, 1]^P$.

Lemma 4.2 (Synchronization identity). *For real $r_p, r_{p'}, s_p, s_{p'}$ write $\xi(s) = 1 - s + s \operatorname{Re} z$ and $\eta(s) = s \operatorname{Im} z$, so that $\alpha = \xi + i\eta$. Then*

$$\operatorname{Im}(\overline{r_p \operatorname{val}'_z(s_p)} r_{p'} \operatorname{val}'_z(s_{p'})) = \frac{2 r_p r_{p'} |z|^2 \operatorname{Im} z (\xi(s_p) \xi(s_{p'}) + \eta(s_p) \eta(s_{p'})) (s_p - s_{p'})}{|\alpha(s_p)|^4 |\alpha(s_{p'})|^4}.$$

In particular, if $r_p r_{p'} \neq 0$ and $s_p, s_{p'} \in (0, 1)$ are distinct, then $r_p \operatorname{val}'_z(s_p)$ and $r_{p'} \operatorname{val}'_z(s_{p'})$ are linearly independent over $\mathbb{R}$, whatever the signs of r_p and $r_{p'}$.

Proof. Write $\alpha_p = \alpha(s_p)$. By Lemma 4.1(d), $\overline{\operatorname{val}'_z(s_p)} \operatorname{val}'_z(s_{p'}) = |z|^2 (\alpha_p \overline{\alpha_{p'}})^2 / (|\alpha_p|^4 |\alpha_{p'}|^4)$. Moreover $\operatorname{Re}(\alpha_p \overline{\alpha_{p'}}) = \xi_p \xi_{p'} + \eta_p \eta_{p'}$ and $\operatorname{Im}(\alpha_p \overline{\alpha_{p'}}) = \eta_p \xi_{p'} - \xi_p \eta_{p'} = (s_p - s_{p'}) \operatorname{Im} z$, and $\operatorname{Im}(w^2) = 2 \operatorname{Re} w \operatorname{Im} w$. For the second statement, $\xi(s) \geq 1 - s > 0$ on $(0, 1)$ since $\operatorname{Re} z \geq 0$, and $\eta_p \eta_{p'} = s_p s_{p'} (\operatorname{Im} z)^2 \geq 0$, so the right-hand side is nonzero; two complex numbers w_1, w_2 are $\mathbb{R}$ -independent exactly when $\operatorname{Im}(\overline{w_1} w_2) \neq 0$. $\square$

Lemma 4.3 (Openness). *Let $r_p, r_{p'} \neq 0$, let $s_p \neq s_{p'}$ lie in $(0, 1)$ and let $\gamma \in \mathbb{C}$. For the map $F(x, y) = \gamma + r_p \operatorname{val}_z(x) + r_{p'} \operatorname{val}_z(y)$ from $\mathbb{R}^2$ to $\mathbb{C}$, the image of every neighbourhood of $(s_p, s_{p'})$ is a neighbourhood of $F(s_p, s_{p'})$.*

Proof. F is continuously differentiable, and its derivative at $(s_p, s_{p'})$, $(x, y) \mapsto r_p \operatorname{val}'_z(s_p)x + r_{p'} \operatorname{val}'_z(s_{p'})y$, is an $\mathbb{R}$ -linear bijection $\mathbb{R}^2 \rightarrow \mathbb{C}$ by Lemma 4.2. The inverse function theorem applies. $\square$

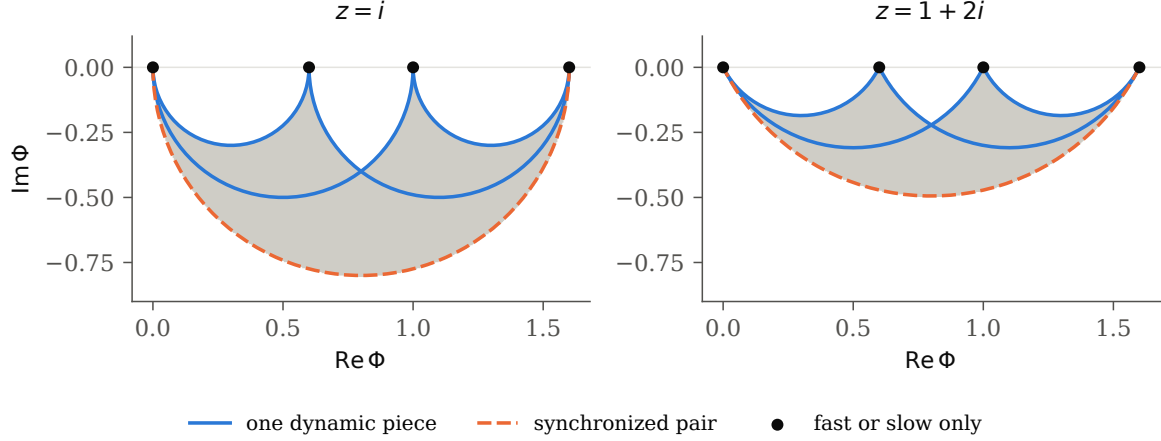


Figure 1: One channel at a fixed eigenvalue z (left: $z = i$; right: $z = 1 + 2i$) with two pieces of loads $r_1 = 1$ and $r_2 = 0.6$. Grey: the attainable channel values $r_1 \text{val}_z(s_1) + r_2 \text{val}_z(s_2)$, $s \in [0, 1]^2$. Coloured: values of boundary-type configurations, in which the pieces are fast, slow, or share one parameter. By Lemma 4.3, configurations with two unequal interior parameters give interior points, so the boundary of the grey region consists of boundary-type values.

5 The exit lemma and non-real eigenvalues

Let A be a star with all loads nonzero, and fix z as in Section 4. For $s \in [0, 1]^P$ define the channel values

$$\Phi_c(s) = \sum_{\text{ch}(p)=c} r_p \text{val}_z(s_p), \quad c \in C.$$

A pair $(q, s) \in (0, \infty)^n \times [0, 1]^P$ is a *generalized contact* if $\det M_q(\Phi(s)) = 0$. A configuration s is *of boundary type in channel c* if all its interior parameters in c coincide: $s_p = s_{p'}$ whenever $\text{ch}(p) = \text{ch}(p') = c$ and $s_p, s_{p'} \in (0, 1)$.

Lemma 5.1 (Exit lemma). *Let (q, s) be a generalized contact and $c \in C$. There are $\kappa \geq 1$ and $s' \in [0, 1]^P$, equal to s outside channel c and of boundary type in channel c , such that $(\kappa q, s')$ is a generalized contact.*

Proof. Let $\hat{\ell}$ be $\Phi(s)$ with its c -th entry replaced by 0. For configurations $\tilde{s}$ that agree with s outside channel c , Lemma 3.2 gives

$$\det M_{\kappa q}(\Phi(\tilde{s})) = a(\kappa) + \Phi_c(\tilde{s}) b(\kappa), \quad a(\kappa) = \det M_{\kappa q}(\hat{\ell}), \quad b(\kappa) = \beta(M_{\kappa q}(\hat{\ell}), u_c, v_c),$$

and a, b are polynomials in κ . Let $K = \sum_p |r_p|$; by Lemma 4.1(c) every channel value of every configuration has modulus at most K , so Lemma 3.3(a) provides $\kappa_0 \geq 1$ such that no generalized contact of this form exists at dilations $\kappa \geq \kappa_0$. The function val_z is continuous, so the set

$$\mathcal{S} = \{(\kappa, \tilde{s}) \in [1, \kappa_0] \times [0, 1]^P : \tilde{s} = s \text{ outside } c, a(\kappa) + \Phi_c(\tilde{s})b(\kappa) = 0\}$$

is closed in a compact set; it contains $(1, s)$. Let $(\kappa_*, s_*) \in \mathcal{S}$ maximize κ .

If $b(\kappa_*) = 0$, then $a(\kappa_*) = 0$. The configuration s' equal to s outside c and to 1 on channel c has $\Phi_c(s') = 0$ by Lemma 4.1(a), so $(\kappa_* q, s')$ is a generalized contact, and s' is of boundary type in c .

If $b(\kappa_*) \neq 0$, then $b \neq 0$ on a neighbourhood of κ_* , where $\psi(\kappa) = -a(\kappa)/b(\kappa)$ is continuous with $\psi(\kappa_*) = \Phi_c(s_*)$. Suppose that s_* is not of boundary type in c : some pieces $p \neq p'$ of channel c have parameters $s_{*p} \neq s_{*p'}$ in $(0, 1)$. By Lemma 4.3, the values $\Phi_c(\tilde{s})$, where $\tilde{s}$ arises from s_* by moving s_{*p} and $s_{*p'}$ inside $(0, 1)$, fill a neighbourhood of $\Phi_c(s_*)$. Hence for some

$\kappa_1 > \kappa_*$ close to κ_* there is such an $\tilde{s}$ with $\Phi_c(\tilde{s}) = \psi(\kappa_1)$, that is, $a(\kappa_1) + \Phi_c(\tilde{s})b(\kappa_1) = 0$. By the choice of κ_0 we have $\kappa_1 < \kappa_0$, and $\kappa_1 > \kappa_* \geq 1$, so $(\kappa_1, \tilde{s}) \in \mathcal{S}$, contradicting the maximality of κ_* . Therefore $s' = s_*$ is of boundary type in c , and $\kappa = \kappa_*$ works. $\square$

Proposition 5.2 (Channel induction). *If (q, s) is a generalized contact, there is a generalized contact (q', s') with s' of boundary type in every channel.*

Proof. Apply Lemma 5.1 to the channels one at a time, in any order. Each step changes only the parameters of its own channel and dilates the core time constants. Being of boundary type in a channel depends only on that channel's parameters, so the channels already treated keep the property. $\square$

No stability hypothesis on B , or on any matrix met along the way, has been used.

Lemma 5.3 (Boundary configurations are boundary systems). *Let $s \in [0, 1]^P$ be of boundary type in every channel. Put $\sigma(p) = \mathbf{F}$ if $s_p = 0$, $\sigma(p) = \mathbf{S}$ if $s_p = 1$, and $\sigma(p) = \mathbf{D}$ otherwise. For each channel c whose group is nonempty let s_c^0 be the common interior parameter and $\tau_c = s_c^0/(1 - s_c^0)$; otherwise $\tau_c = 1$. Then for every q ,*

$$M_q(\Phi(s)) = zQ - B_\sigma - \sum_c \frac{R_c^\sigma}{1 + z\tau_c} u_c v_c^\top,$$

the reduced pencil of Bd_σ at time constants (q, τ) . Consequently, if (q, s) is a generalized contact, then z is an eigenvalue of $\text{Bd}_\sigma \text{diag}(q, \tau)^{-1}$.

Proof. By Lemma 4.1(a),(b), $\Phi_c(s) = \sum_{\text{ch}(p)=c, \sigma(p)=\mathbf{F}} r_p + R_c^\sigma/(1 + z\tau_c)$, and the fast terms are exactly the static loads in B_σ . Since $\text{Re } z \geq 0$ and $\tau_c > 0$, $1 + z\tau_c \neq 0$, and Lemma 3.1, applied to Bd_σ , gives the last statement. $\square$

Proof of Theorem 2.3 for non-real z . Let z with $\text{Re } z \geq 0$ and $\text{Im } z \neq 0$ be an eigenvalue of $A \text{diag}(q, t)^{-1}$. Since $\text{Re } z \geq 0$, $1 + zt_p \neq 0$, and Lemma 3.1 shows that $M_q(\ell(z))$ is singular. By Lemma 4.1(b), $\ell(z) = \Phi(s)$ with $s_p = t_p/(1 + t_p)$, so (q, s) is a generalized contact. Proposition 5.2 and Lemma 5.3 finish the proof. $\square$

6 Real eigenvalues, lifting and the criterion

Lemma 6.1 (Zero eigenvalues). *If 0 is an eigenvalue of $A \text{diag}(q, t)^{-1}$ for a star A , then 0 is an eigenvalue of a positive scaling of $\text{Bd}_{\sigma \equiv \mathbf{F}}$ with the same core time constants q .*

Proof. At $z = 0$, $\ell_c(0) = \sum_{\text{ch}(p)=c} r_p$ and $M_q(\ell(0)) = -B_{\sigma \equiv \mathbf{F}}$. The boundary system $\text{Bd}_{\sigma \equiv \mathbf{F}}$ has only zero lumps, so its reduced pencil at $z = 0$ is also $-B_{\sigma \equiv \mathbf{F}}$. Apply Lemma 3.1 twice. $\square$

Lemma 6.2 (Real positive eigenvalues). *If a positive scaling $A \text{diag}(q, t)^{-1}$ of a star has a real eigenvalue $z > 0$, then there are a boundary choice σ without fast pieces and $\tau \in (0, \infty)^C$ such that z is an eigenvalue of $\text{Bd}_\sigma \text{diag}(q, \tau)^{-1}$.*

Proof. Put $a_p = 1/(1 + zt_p) \in (0, 1)$, so that the channel values $\ell_c = \ell_c(z) = \sum_{\text{ch}(p)=c} r_p a_p$ are real, and let G_c^+ and G_c^- be the sums of the positive and of the negative loads of channel c . If $\ell_c > 0$, some load of c is positive, and $\ell_c \leq \sum_{\text{ch}(p)=c, r_p > 0} r_p a_p < G_c^+$; symmetrically $G_c^- < \ell_c$ if $\ell_c < 0$. Define $\sigma(p) = \mathbf{D}$ if $\ell_{\text{ch}(p)} \neq 0$ and $r_p \ell_{\text{ch}(p)} > 0$, and $\sigma(p) = \mathbf{S}$ otherwise. Then $B_\sigma = B$, and R_c^σ equals G_c^+ , G_c^- or 0 according as ℓ_c is positive, negative or zero. Put $\tau_c = (R_c^\sigma/\ell_c - 1)/z > 0$ if $\ell_c \neq 0$ and $\tau_c = 1$ otherwise. Then $R_c^\sigma/(1 + z\tau_c) = \ell_c$ for every c , so the reduced pencil of Bd_σ at (q, τ) coincides with $M_q(\ell(z))$, which is singular by Lemma 3.1. Apply Lemma 3.1 to Bd_σ . $\square$

Proof of Theorem 2.3. The cases $z = 0$, $z > 0$ and $\text{Im } z \neq 0$ are Lemma 6.1, Lemma 6.2 and Section 5. $\square$

Proof of Theorem 2.4. Let z_0 , $\text{Re } z_0 > 0$, be an eigenvalue of $\text{Bd}_\sigma \text{diag}(q, \tau)^{-1}$. For $\varepsilon \in [0, 1]$ and $\text{Re } z > 0$ put

$$\Delta_\varepsilon(z) = \det\left(zQ - B - \sum_c \ell_c^\varepsilon(z) u_c v_c^\top\right), \quad \ell_c^\varepsilon(z) = \sum_{\substack{\text{ch}(p)=c \\ \sigma(p)=F}} \frac{r_p}{1+z\varepsilon} + \sum_{\substack{\text{ch}(p)=c \\ \sigma(p)=D}} \frac{r_p}{1+z\tau_c} + \sum_{\substack{\text{ch}(p)=c \\ \sigma(p)=S}} \frac{r_p \varepsilon}{\varepsilon+z}.$$

For $\varepsilon > 0$ this is the reduced determinant (1) of A at the time constants q (core), ε (F-pieces), $\tau_{\text{ch}(p)}$ (D-pieces) and $1/\varepsilon$ (S-pieces), since $\varepsilon/(\varepsilon+z) = 1/(1+z/\varepsilon)$. At $\varepsilon = 0$ it is the reduced determinant of Bd_σ at (q, τ) , so $\Delta_0(z_0) = 0$ by Lemma 3.1. All denominators have positive real part, so $(\varepsilon, z) \mapsto \Delta_\varepsilon(z)$ is continuous on $[0, 1] \times \{\text{Re } z > 0\}$ and holomorphic in z . Each $|\ell_c^0(z)| \leq \sum_p |r_p|$, so by Lemma 3.3(b) $\Delta_0(z) \neq 0$ for all large real z . Hence the zeros of Δ_0 in the right half-plane are isolated, and there is $\varrho \in (0, \text{Re } z_0/2)$ such that Δ_0 has no zero on the circle $|z - z_0| = \varrho$. Let $\delta > 0$ be the minimum of $|\Delta_0|$ on this circle. By uniform continuity on the compact set $[0, 1] \times \{|z - z_0| \leq \varrho\}$ there is $\varepsilon \in (0, 1]$ with $|\Delta_\varepsilon - \Delta_0| < \delta/2$ on the closed disc. If Δ_ε had no zero in the closed disc, the maximum modulus principle applied to $1/\Delta_\varepsilon$ would give $|\Delta_\varepsilon(z_0)| \geq \min_{|z-z_0|=\varrho} |\Delta_\varepsilon| \geq \delta/2$, whereas $|\Delta_\varepsilon(z_0)| = |\Delta_\varepsilon(z_0) - \Delta_0(z_0)| < \delta/2$. So $\Delta_\varepsilon(z_1) = 0$ for some z_1 with $\text{Re } z_1 \geq \text{Re } z_0 - \varrho > 0$, and z_1 is an eigenvalue of the corresponding positive scaling of A by Lemma 3.1. (This is Hurwitz's theorem on zeros of uniform limits; see [11, Chapter VII]. The eigenvalue z_1 is close to z_0 but in general different from it.) $\square$

Proof of Corollary 2.5. (a) If some positive scaling of A had an eigenvalue z with $\text{Re } z \geq 0$, Theorem 2.3 would give the same eigenvalue for a positive scaling of some Bd_σ , which is impossible if Bd_σ is D-stable. (b) If A is D-unstable, Theorem 2.3 with $\text{Re } z > 0$ shows that some Bd_σ is D-unstable; the converse is Theorem 2.4. $\square$

Proof of Proposition 2.6. Let the time constants be q_1, q_2 (core) and t (piece), and let $\ell = 1/(1+zt)$. The reduced pencil and its determinant are

$$\begin{pmatrix} zq_1 + 1 - \ell & 1 \\ -1 & zq_2 \end{pmatrix}, \quad (zq_1 + 1 - \ell)zq_2 + 1.$$

Suppose that it vanishes for some z with $\text{Re } z \geq 0$. Then $z \neq 0$, and $zq_1 + 1 - \ell = -1/(zq_2)$, whose real part is $-\text{Re } z/(q_2|z|^2) \leq 0$. On the other hand $\text{Re } \ell = (1+t\text{Re } z)/|1+zt|^2 < 1$, because $|1+zt|^2 = (1+t\text{Re } z)^2 + t^2(\text{Im } z)^2 > 1+t\text{Re } z$ for $z \neq 0$; hence $\text{Re}(zq_1 + 1 - \ell) = q_1 \text{Re } z + 1 - \text{Re } \ell > 0$, a contradiction. By Lemma 3.1, no positive scaling of A has an eigenvalue in the closed right half-plane, so A is D-stable. The boundary system with the piece fast has a zero lump, and its core and its reduced pencil at $z = i$ and unit time constants are

$$B + \mathbf{e}_1 \mathbf{e}_1^\top = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad iI - (B + \mathbf{e}_1 \mathbf{e}_1^\top) = \begin{pmatrix} i & 1 \\ -1 & i \end{pmatrix}.$$

The latter annihilates $(1, -i)$, so i is an eigenvalue by Lemma 3.1. $\square$

The other two boundary systems of this example are D-stable: with the piece slow the core is B , whose scalings have trace $-d_1 < 0$ and determinant $d_1 d_2 > 0$, and with the piece synchronized the boundary system is A itself. So the failure in Proposition 2.6 comes from a single unattained limit: the quasi-steady model is a conservative oscillator, while no finite time constant of the piece realizes it.

7 Sharpness

The following inequality is the secant condition of Tyson and Othmer [31] and Thron [30] (see also [27, 4]), stated on the closed right half-plane. We include the short proof, since the closed half-plane is needed.

Lemma 7.1 (Secant inequality). *Let $N \geq 1$, $\tau_1, \dots, \tau_N > 0$, $g > 0$, and $\lambda \in \mathbb{C}$ with $\operatorname{Re} \lambda \geq 0$ and $\prod_{j=1}^N (1 + \lambda \tau_j) = -g$. Then $N \geq 3$ and $g \cos^N(\pi/N) \geq 1$.*

Proof. Write $1 + \lambda \tau_j = |1 + \lambda \tau_j| e^{i\theta_j}$ with $|\theta_j| < \pi/2$; this is possible since $\operatorname{Re}(1 + \lambda \tau_j) \geq 1$, which also gives $|1 + \lambda \tau_j| \cos \theta_j \geq 1$. The product condition gives $\prod |1 + \lambda \tau_j| = g$ and $\sum \theta_j \in \pi + 2\pi\mathbb{Z}$, so $\sum |\theta_j| \geq \pi$; since each $|\theta_j| < \pi/2$, $N \geq 3$. The function $\log \cos$ is concave on $[0, \pi/2)$ and $\cos$ is decreasing there, so with $\bar{\theta} = \frac{1}{N} \sum |\theta_j| \geq \pi/N$,

$$1 \leq \prod_j |1 + \lambda \tau_j| \cos \theta_j = g \prod_j \cos |\theta_j| \leq g \cos^N \bar{\theta} \leq g \cos^N(\pi/N). \quad \square$$

Lemma 7.2 (Monotonicity). *$N \mapsto \cos^N(\pi/N)$ is strictly increasing on $N \geq 3$.*

Proof. $\cos^N(\pi/N) = \exp(-\pi f(\pi/N))$ with $f(x) = -\log \cos(x)/x$. Since $-\log \cos$ is strictly convex on $(-\pi/2, \pi/2)$ and vanishes at 0, its chord slope f is strictly increasing on $(0, \pi/2)$. $\square$

Proof of Theorem 2.7. Write $\varsigma_c = 1$ for $c \neq k-1$ and $\varsigma_{k-1} = -1$, so that $u_c = \varsigma_c \mathbf{e}_c$, and write $\sigma(c)$ for the role of the unique piece of channel c .

Instability. At unit time constants let $\zeta = e^{i\pi/k}$, $\mu = \sqrt{\rho} e^{i\pi/2k}$, $x_c = \zeta^c$ and $w_c = x_{c+1}/\mu$ (indices mod k). The piece rows $x_{c+1} - w_c = (\mu - 1)w_c$ hold by definition. The core rows read $-x_c + \rho \varsigma_c w_c = (\mu - 1)x_c$, that is, $\rho \varsigma_c x_{c+1} = \mu^2 x_c = \rho \zeta x_c$, which holds for $c < k-1$ since $x_{c+1} = \zeta x_c$, and for $c = k-1$ since $-x_0 = -1 = \zeta^k = \zeta x_{k-1}$. So $\mu - 1$ is an eigenvalue, and $\operatorname{Re}(\mu - 1) = \sqrt{\rho} \cos(\pi/2k) - 1 > 0$ by (2).

Boundary systems with fewer than k groups. Let σ be such a choice, so some channel c_* has $\sigma(c_*) \neq D$. Suppose that z , with $\operatorname{Re} z \geq 0$, is an eigenvalue of $\operatorname{Bd}_\sigma \operatorname{diag}(q, \tau)^{-1}$. By Lemma 3.1 there is $x \neq 0$ with $(1 + zq_c)x_c = \varsigma_c L_c x_{c+1}$ for all c , where $L_c = \rho$ if $\sigma(c) = F$, $L_c = 0$ if $\sigma(c) = S$, and $L_c = \rho/(1 + z\tau_c)$ if $\sigma(c) = D$. Since $1 + zq_c \neq 0$, a zero x_{c+1} forces $x_c = 0$, and going around the cycle would give $x = 0$; so no x_c vanishes and no channel is slow. Multiplying the relations around the cycle,

$$\prod_c (1 + zq_c) \prod_{\sigma(c)=D} (1 + z\tau_c) = -\rho^k,$$

a product of $N = k + |\{c : \sigma(c) = D\}| \leq 2k - 1$ factors of the form $1 + z\tau$ with $\tau > 0$. By Lemmas 7.1 and 7.2, $1 \leq \rho^k \cos^N(\pi/N) \leq \rho^k \cos^{2k-1}(\pi/(2k-1)) < 1$, a contradiction.

The window. By Lemma 7.2, $\cos^{2k-1}(\pi/(2k-1)) < \cos^{2k}(\pi/2k)$, so some ρ satisfies $\sec^{2k}(\pi/2k) < \rho^k < \sec^{2k-1}(\pi/(2k-1))$; the first inequality is equivalent to $\rho \cos^2(\pi/2k) > 1$. $\square$

For Proposition 2.8 we use the Routh–Hurwitz condition for cubics, in the following elementary form.

Lemma 7.3 (Cubics). *Let $a_3, a_2, a_1, a_0 > 0$ with $a_2 a_1 > a_3 a_0$. Then $a_3 z^3 + a_2 z^2 + a_1 z + a_0 \neq 0$ whenever $\operatorname{Re} z \geq 0$.*

Proof. Let $z = x + iy$ with $x \geq 0$ be a root. If $y = 0$, the polynomial is positive at x . Otherwise the imaginary part, divided by y , gives $a_3 y^2 = 3a_3 x^2 + 2a_2 x + a_1$, and substituting this into a_3 times the real part gives $0 = -8a_3^2 x^3 - 8a_2 a_3 x^2 - 2a_1 a_3 x - 2a_2^2 x + (a_0 a_3 - a_1 a_2) < 0$. $\square$

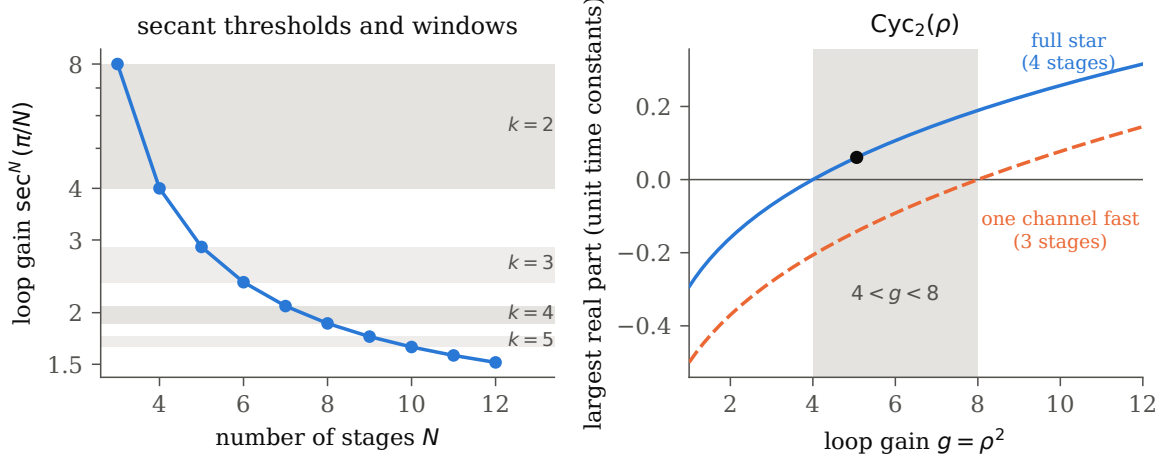


Figure 2: Left: the secant thresholds $\sec^N(\pi/N)$; for k channels the window of Theorem 2.7 lies between $N = 2k$ and $N = 2k - 1$ stages (shaded for $k = 2, \dots, 5$). Right: for Cyc_2 , the largest real part at unit time constants of the full star (a 4-stage cycle) and of the boundary systems with one fast channel (3-stage cycles), as functions of the loop gain $g = \rho^2$. In the window $4 < g < 8$ only the star itself is unstable; the dot marks Example 7.4, $g = 81/16$.

Proof of Proposition 2.8. Instability. At unit time constants let $\mu = c e^{i\pi/3} = \frac{c}{2} + \frac{c\sqrt{3}}{2}i$, so that $\mu^3 = -c^3$. The vector with $x = (\mu^2, \mu)$ and $w_p = 1$ for every p is an eigenvector for $\mu - 1$: the rows read $-\mu^2 - \frac{c^3}{m} \sum_p w_p = -\mu^2 + \mu^3 = (\mu - 1)\mu^2$, $\mu^2 - \mu = (\mu - 1)\mu$ and $\mu - 1 = (\mu - 1)w_p$. The real part of $\mu - 1$ is $c/2 - 1 > 0$.

Boundary systems. Let σ synchronize fewer than m pieces, and let X and R be its fast and synchronized load sums, so that $X \geq 0$ and $0 \leq R \leq (m-1)c^3/m < 8$. At time constants (q, τ) the reduced pencil of Bd_σ is $\begin{pmatrix} 1+zq_1 & X+R/(1+z\tau) \\ -1 & 1+zq_2 \end{pmatrix}$. If it is singular for some z with $\text{Re } z \geq 0$, multiplying its determinant by $1 + z\tau \neq 0$ gives $a_3 z^3 + a_2 z^2 + a_1 z + a_0 = 0$ with

$$a_3 = q_1 q_2 \tau, \quad a_2 = q_1 q_2 + q_1 \tau + q_2 \tau, \quad a_1 = q_1 + q_2 + \tau + X\tau, \quad a_0 = 1 + X + R.$$

All coefficients are positive, and

$$a_2 a_1 - a_3 a_0 = q_1(q_2 - \tau)^2 + q_2(q_1 - \tau)^2 + \tau(q_1 - q_2)^2 + (8 - R)q_1 q_2 \tau + X\tau^2(q_1 + q_2) > 0,$$

which contradicts Lemma 7.3. By Lemma 3.1, Bd_σ is D-stable. $\square$

Example 7.4 ($k = 2$). Take $\rho = 9/4$, so $g = 81/16$ and (2) reads $9/8 > 1$ and $81/128 < 1$. At unit time constants the characteristic polynomial of $\text{Cyc}_2(9/4)$ is $(\lambda + 1)^4 + 81/16$, whose roots are $-1 + \frac{3}{2}e^{\pm i\pi/4}$ and $-1 + \frac{3}{2}e^{\pm 3i\pi/4}$. The first pair has real part $\frac{3\sqrt{2}}{4} - 1 \approx 0.0607 > 0$, and the Routh array has two sign changes. The nine boundary systems are listed in Table 2. A slow channel cuts the cycle. Two fast channels leave a 2-stage cycle on the core, with matrix $-I + \rho(\mathbf{e}_0 \mathbf{e}_1^\top - \mathbf{e}_1 \mathbf{e}_0^\top)$, which is D-stable for every ρ . One fast and one synchronized channel leave a 3-stage cycle of gain $81/16 < 8 = \sec^3(\pi/3)$. Only the fully synchronized choice, which is the star itself, is unstable.

8 Discussion

What the theorems say for reduced models. Theorem 2.3 says that every way in which a star can reach or cross the imaginary axis, for some choice of time constants, is

Table 2: Boundary systems of $\text{Cyc}_2(9/4)$; $\sigma = (\sigma(0), \sigma(1))$, loop gain $g = 81/16$. “Stages” counts the first-order stages left on the feedback cycle.

σ	reduced model	stages	verdict	reason
$(S, \cdot), (\cdot, S)$	a frozen piece	cycle cut	D-stable	triangular after permutation
(F, F)	both pieces quasi-steady	2	D-stable	2×2 : trace < 0 , det > 0
$(F, D), (D, F)$	one piece quasi-steady	3	D-stable	$g < \sec^3(\pi/3) = 8$
(D, D)	the full star	4	D-unstable	$g > \sec^4(\pi/4) = 4$

already present, at the same eigenvalue, in a reduced model built by quasi-steady elimination, freezing and lumping, with at most one lumped pool per interface (possibly at other core time constants). Theorem 2.4 says that nothing is lost in the other direction for strict instability: a reduced model with a growing mode certifies a growing mode of the full model, at a nearby eigenvalue, when the lumped pools keep their time constants, the eliminated pieces are fast enough and the frozen pieces slow enough. Theorem 2.7 and Proposition 2.8 show that the lumped pools cannot be dropped: in Cyc_k every reduced model with a quasi-steady or frozen channel is D-stable, and the instability appears only with all k channels dynamic; and in the example of Proposition 2.8 it appears only when all m pieces are lumped.

The marginal case. Corollary 2.5 decides D-semistability exactly, but D-stability only in one direction. When all boundary systems are D-semistable and some are marginal, the star is D-semistable, and it is D-stable exactly when no positive scaling of it has an eigenvalue on the imaginary axis. Theorem 2.3 transfers such eigenvalues to boundary systems but not back, and Proposition 2.6 shows that the transfer back can fail. For two boundary systems it always works. If $\det B_{\sigma \equiv F} = 0$, then $M_q(\ell(0)) = -B_{\sigma \equiv F}$ is singular for all q , so by Lemma 3.1 every positive scaling of A has the eigenvalue 0. And if z , with $\text{Re } z \geq 0$, is an eigenvalue of $\text{Bd}_{\sigma \equiv D} \text{diag}(q, \tau)^{-1}$, then giving every piece of channel c the time constant τ_c makes the reduced pencil of A equal to that of $\text{Bd}_{\sigma \equiv D}$, so z is an eigenvalue of that scaling of A by Lemma 3.1. Hence a D-stable star has $\det B_{\sigma \equiv F} \neq 0$ and a D-stable fully synchronized boundary system, and the gap concerns the other boundary systems. For one coordinate channel with positive loads, a D-stable core and $n \geq 2$, the companion paper [12, Lemma 6.1] closes this gap: the star is D-stable if and only if $\det(-B_{\sigma \equiv F}) > 0$ and every boundary system with at most one dynamic piece, other than the purely static $\text{Bd}_{\sigma \equiv F}$, is D-stable. Proposition 2.6 is an instance, in which the purely static system is the only marginal one.

Size of the reduction. The number of boundary choices is 3^m , but by Remark 2.9(iii) only the per-channel pairs of fast and synchronized load sums matter. If the loads are $r_p = \delta n_p$ with $\delta \in \mathbb{R}$ and integers $n_p \geq 0$, these pairs take at most $(N_c + 1)^2$ values in channel c , where $N_c = \sum_{\text{ch}(p)=c} n_p$, so the star has at most $\prod_c (N_c + 1)^2$ distinct boundary systems (Section 9). For a fixed number of channels this is polynomial in the loads written in unary; for instance, twenty equal loads in one channel give $3^{20} \approx 3.5 \cdot 10^9$ boundary choices but at most 441 distinct boundary systems. For binary-encoded loads, deciding D-stability of stars is coNP-hard, and deciding D-instability NP-hard, already for one coordinate channel and a 4×4 core with instance-dependent entries of polynomial bit length [14]. So, unless $P = NP$, D-stability of stars cannot be decided in time polynomial in the binary input, even though every boundary system of those instances has order five.

Open problems.

- (1) *Coordinate channels.* The channels of Cyc_k read one coordinate and act on another. For coordinate channels, $u_c = v_c = \mathbf{e}_{i_c}$, with positive loads, it is open whether two channels

can require two synchronized groups.

- (2) *The marginal case* for general interfaces: which additional reduced information decides D-stability when all boundary systems are D-semistable, $\det B_{\sigma \equiv F} \neq 0$ and $\text{Bd}_{\sigma \equiv D}$ is D-stable, but some other boundary system is marginal?
- (3) *Higher-order attachments*. For attachments of higher order the value of a module at a fixed eigenvalue ranges over a two-dimensional region rather than an arc, and the boundary structure of Minkowski sums of such regions replaces Lemma 4.2.
- (4) *Nonlinear dynamics*. The results are linear and spectral; they say nothing directly about the nonlinear onset of oscillations.

9 Formal verification

The results are formally verified in Lean 4 (v4.30.0) with Mathlib at revision c5ea00351c28 (the full hash is in the ancillary Lake manifest). The development consists of the definition file `proofs/DUnstableCores/DScaling.lean` and sixteen modules in `proofs/DStabilityLocalization/`: thirteen for the main development, `GroupSize` (Proposition 2.8 for every m), `Counting` (Remark 2.9(iii) and the counting bound of Section 8), and the root file `Paper.lean`, which imports all of them. The definition file defines positive scalings (`rightScale`), eigenpairs (`HasEigenpair`) and the predicates `DStable`, `DNonUnstable` (D-semistable) and `DUnstable` exactly as in Section 2; an eigenpair is a nonzero complex vector y with $(AD)y = zy$.

The strict compilation receipt of `Paper.lean` (warnings treated as errors) exports the following declarations.

- `localization_and_sharpness` states, for all finite index types in the universe `Type`: Theorem 2.4 for arbitrary loads; Theorem 2.3 and Corollary 2.5 for nonzero loads; Theorem 2.7 as printed, for every $k = j + 2$ with $j \in \mathbb{N}$ (explicit eigenvalue, instability, D-stability of the boundary systems with fewer than k groups for every ρ satisfying (2), and existence of such ρ); Proposition 2.6 as printed; Proposition 2.8 for $m = 2$ and $c = 5/2$; and the verdicts of Example 7.4.
- `group_size_sharpness` states Proposition 2.8 as printed, for every $m = j + 2$ with $j \in \mathbb{N}$: nonzero loads, the explicit eigenvalue and its real part $c/2 - 1 > 0$, D-instability, D-stability of every boundary system in which fewer than m pieces are synchronized, and the existence of an admissible c .
- `bdSystem_eq_of_loads` and `card_bdSystems_le` state Remark 2.9(iii) and the bound $\prod_c (N_c + 1)^2$ of Section 8.

All of them depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`; no `sorry`, custom axiom or native computation is used. The lemmas are declarations on which these depend, with the statements printed here; in particular Lemma 5.3 exhibits the scaling $\text{diag}(q, \tau)^{-1}$, and Lemmas 6.1 and 6.2 keep the core time constants, as in Remark 2.9(i). Table 3 lists the declarations.

The other remarks, the modelling reading of Table 1 and the observations of Section 8 are not formalized; the complexity statements of Section 8 are quoted from [14]. The characteristic polynomial, roots and Routh array of Example 7.4, the Hurwitz data behind Table 2 and Proposition 2.8, the values plotted in the figures, brute-force checks of the counting bound and of the two observations on the marginal case in small examples, and the source hashes of the compilation receipt are replayed exactly by the script `check_paper.py`. The ancillary material consists of the Lean sources with a Lake configuration pinned to the Mathlib revision above, a path-free summary of the receipt, the replay script and the two companion manuscripts.

Table 3: Lean declarations (namespace `DStabilityLocalization`); all are imported by `Paper.lean`.

Result	Declarations
Definitions 2.1, 2.2	<code>star</code> , <code>pencil</code> , <code>lagValue</code> , <code>Role</code> , <code>bdCore</code> , <code>bdLoad</code> , <code>bdSystem</code> , <code>groups</code>
Lemma 3.1	<code>pencil_kernel_of_eigenpair</code> , <code>eigenpair_of_pencil_kernel</code>
Lemma 3.2	<code>det_sub_smul_vecMulVec</code>
Lemma 3.3	(a) <code>pencil_det_ne_of_large</code> , (b) <code>pencil_det_ne_of_norm</code>
Lemma 4.1	<code>val</code> , <code>val_zero</code> , <code>val_one</code> , <code>lag_eq_val</code> , <code>val_eq_lag</code> , <code>norm_val_le_one</code> , <code>hasStrictDerivAt_val</code> , <code>dval_eq</code>
Lemma 4.2	<code>sync_det</code> , <code>sync_det_ne</code>
Lemma 4.3	<code>open_two_pieces</code>
Lemma 5.1	<code>exit_channel</code>
Proposition 5.2	<code>localize_all</code>
Lemma 5.3	<code>pencil_boundary</code> , <code>boundary_eigenpair</code>
Theorem 2.3, non-real z	<code>localization_nonreal</code>
Lemmas 6.1, 6.2	<code>zero_eigen_static</code> , <code>localization_real</code>
Theorem 2.3	<code>spectral_localization</code>
Theorem 2.4	<code>dUnstable_of_boundary</code> , <code>with_hurwitz_zero</code> , <code>exists_sphere_ne</code>
Corollary 2.5	<code>dStable_of_boundary</code> , <code>dNonUnstable_iff</code> , <code>dUnstable_iff</code>
Proposition 2.6	<code>gap_star_dStable</code> , <code>gap_static_eigenpair</code> , <code>marginal_gap</code>
Lemmas 7.1, 7.2	<code>secant_inequality</code> , <code>cos_pow_strictMono</code>
Theorem 2.7	<code>cycStar</code> , <code>cycEig</code> , <code>cyc_star_eigenpair</code> , <code>cycEig_re_pos</code> , <code>cyc_star_dUnstable</code> , <code>cyc_lower_dStable</code> , <code>cyc_window</code> , <code>sharpness_full</code>
Lemma 7.3	<code>cubic_ne_zero</code>
Proposition 2.8	<code>grp_star_eigenpair</code> , <code>grp_lower_dStable</code> , <code>grp_window</code> , <code>group_size_sharpness</code> ; for $m = 2$: <code>lump_star_eigenpair</code> , <code>lump_lower_dStable</code> , <code>lumping_needed</code>
Example 7.4, Table 2	<code>cyc2_example</code>
Remark 2.9(iii), Section 8	<code>bdSystem_eq_of_loads</code> , <code>card_bdSystems_le</code>
Main results	<code>localization_and_sharpness</code> , <code>group_size_sharpness</code>

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